Evaluating Detection Models

Review Learning Objectives

Submit your assignment

Due Mar 3, 11:59 PM IST

Try again

Receive grade

To Pass 75% or higher

Your grade
100%

View Feedback
We keep your highest score

Congratulations! You passed!

Grade received 100% **Latest Submission Grade** 100% **To pass** 75% or higher

Go to next item

1. True or false: classification models typically require more computational resources and data than detection models. **1 / 1 point**

- ○ True
- ◉ False

Correct
Detection models usually require more resources and data than classification.

2. Assume you've trained a model, but the mean average precision does not meet your requirements. When analyzing your labeled data, you find that some classes are well separated into clusters based on area and aspect ratio. You also see that the range in object areas spans a factor of 50. **1 / 1 point**

What approach below is likely to help improve your mean average precision the most?

- ○ Increase the mini-batch size to ensure all classes are represented.
- ○ Acquire more data.
- ◉ Train multiple models based on the clusters of classes.

Correct
This is a good approach for this scenario. The tradeoff is more memory and computational resources.

3. When evaluating your model, you find several classes with an average precision of 1. Which statement below is true about these classes? **1 / 1 point**

- ◉ All objects in the test data for this class are successfully detected with higher confidence scores than any false positives.
- ○ All true positives have a confidence score of 1, so any detection threshold can be used.
- ○ The detector finds all the objects of that class in the test data with no false positives.

Correct
An AP of one means that all false positives have lower confidence than true positives, **and** all objects are detected before there is a false positive.

4. Assume you are using a model that requires anchor boxes and want to improve your model. Which statement below is true? **1 / 1 point**

- ○ Increasing the number of anchor boxes may improve results and your model's prediction speed.
- ◉ Increasing the number of anchor boxes may improve your results at the cost of more computational resources for training and prediction.
- ○ The optimal number of anchor boxes is determined by the `estimateAnchorBoxes` function.
- ○ The number of anchor boxes is not an important hyperparameter.

Correct
Increasing anchor boxes is a good first step to improving results. However, anchor boxes increase model complexity.

Open the file `Week3quiz.mlx` in MATLAB and evaluate the provided fastenerTinyYolo4Detector model following a similar analysis as you did earlier. Then, answer the questions below.

5. Use a detection threshold of 0.05 for detection and an IoU threshold of 0.5 to evaluate the fastener model. What is the mAP? **1 / 1 point**

- ○ 0.51
- ◉ 0.76
- ○ 0.9

Correct
This is the mean of the average precision across the four classes.

6. Using an IoU threshold of 0.5 and detection threshold of 0.05, which class has the lowest average precision? **1 / 1 point**

- ○ Nail
- ○ Nut
- ◉ Screw
- ○ Washer

Correct
The detector struggles detecting screws.

7. Lower the IoU threshold to 0.25. What is the mAP? **1 / 1 point**

- ○ 0.75
- ◉ 0.90
- ○ 0.99

Correct
The IoU increases to 0.9 with the lower IoU threshold.

8. Which statement below most accurately describes the increases in mAP when decreasing the IoU threshold. **1 / 1 point**

- ○ Decreasing the IoU does not help recall. Many objects are still missed.
- ◉ The detector is getting the class predictions correct but struggling with location accuracy.
- ○ Location accuracy is not an issue for this detector.

Correct
The detector evaluation results improve dramatically when decreasing the IoU threshold. This indicates that many class predictions are correct, but location accuracy is an issue for the detector.

9. Think back to your analysis of the ground truth data in Week 2. What statement below is the best explanation for the low-performing classes? **1 / 1 point**

- ◉ The Nail and Screw classes have a wide range of aspect ratios, so the detector struggles more with these classes, especially with location accuracy.
- ○ The Nail and Screw classes look very similar and are mixed up by the detector.
- ○ The Nut and Washer classes are very similar and mixed up by the detector.
- ○ There is a significant class imbalance in the dataset.

Correct
The wide range of bounding box sizes and shapes with these two classes means more data might be necessary.